

Introduction to Probability and Statistics

Lecture 4 Solutions

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Exercise 1: Chapter 9, problem 4 and 9

The following data indicate the relationship between x , the specific gravity of a wood sample, and Y , its maximum crushing strength in compression parallel to the grain

x_i	$y_i(\text{psi})$	x_i	$y_i(\text{psi})$
.41	1850	.39	1760
.46	2620	.41	2500
.44	2340	.44	2750
.47	2690	.43	2730
.42	2160	.44	3120

- Plot a scatter diagram. Does a linear relationship seem reasonable?
- Estimate the regression coefficients.
- Predict the maximum crushing strength of a wood sample whose specific gravity is 0.43.
- Estimate the variance of an individual response.

Solution:

- Just by looking at the scatter plot, a linear model seems reasonable. This is confirmed after looking at the residuals in Fig. 1.
- The regression coefficients are $\beta_0 = -2825.9168$ and $\beta_1 = 12245.7466$
- We estimate the response by calculating $r(0.43) = \beta_0 + \beta_1(0.43) = 2439.7542$
- This is estimated from the residuals as $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2 = 105660.066$

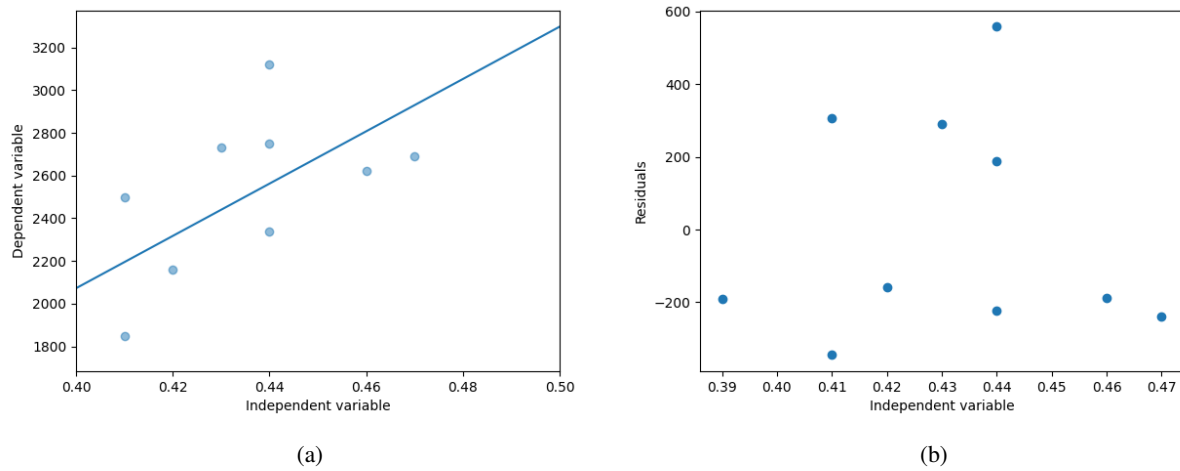


Fig. 1. Linear regression and residual analysis for the crushing strength of wood.

Exercise 2: A measurement campaign involving two wireless devices was performed. One device (receiver) was kept on a fixed position, while the other (transmitter) was moved around and placed at several distances d_i for $i = 1, 2, \dots$ from the receiver. The transmitter was configured to send a known signal toward the receiver in every position and the receiver calculated the received power for every transmission. The file `received_power.csv` contains the $n = 500$ data obtained, in which P_i is the received power, measured in milliWatts [mW], and $d_i \in [30, 300]$ [m] is the distance between the transmitter and receiver.

- Plot a scatter diagram. Does a linear relationship seem reasonable?
- Use linear regression directly to estimate the regression coefficients. Then, compute the coefficient of determination, evaluate the standardized residuals and plot them. Can you describe the residual behavior?
- A telecommunications engineer tells you that the relation between power and distance is

$$P_i = \frac{K}{d_i^\gamma L_i}$$

where K is a constant and L_i represents additional (random) losses. They also suggest to use the decibel [dB] version of the formula, so you should apply $10 \log_{10}$ to each side of the equation before trying the linear regression. How the equation becomes? Can you identify which are the terms $Y_i, X_i, \beta_0, \beta_1, \epsilon_i$ in the new equations?

- d) In the previous question, how should L_i be distributed in order to have ϵ_i satisfying all the assumptions for linear regression? (Tip: There is no need to evaluate the function of the distribution mathematically, just explain which distribution ϵ_i should have and try to discuss how L_i should be.)
- e) Estimate the regression coefficients with the first $n' = 50$ measurements, compute the coefficient of determination, evaluate and plot the standardized residuals.
- f) Estimate the regression coefficients with all the results $n = 500$, compute the coefficient of determination, evaluate and plot the standardized residuals. Comment the results comparing them with the ones obtained in points b) and e).
- g) Give a possible estimate of K and γ based on the regression coefficients β_0 and β_1 . Then, compute them numerically using the regression coefficients computed in e) and f),
- h) Predict the mean response for new data $d_* = 400$ and compute the 95% prediction interval using the regression parameters estimated with $n' = 50$ and $n = 500$. Comment the results. (Tip: Remember that your x_* is not just d_* . You need to apply the transformation!)

Solution:

- a) It is clearly not a linear relation.

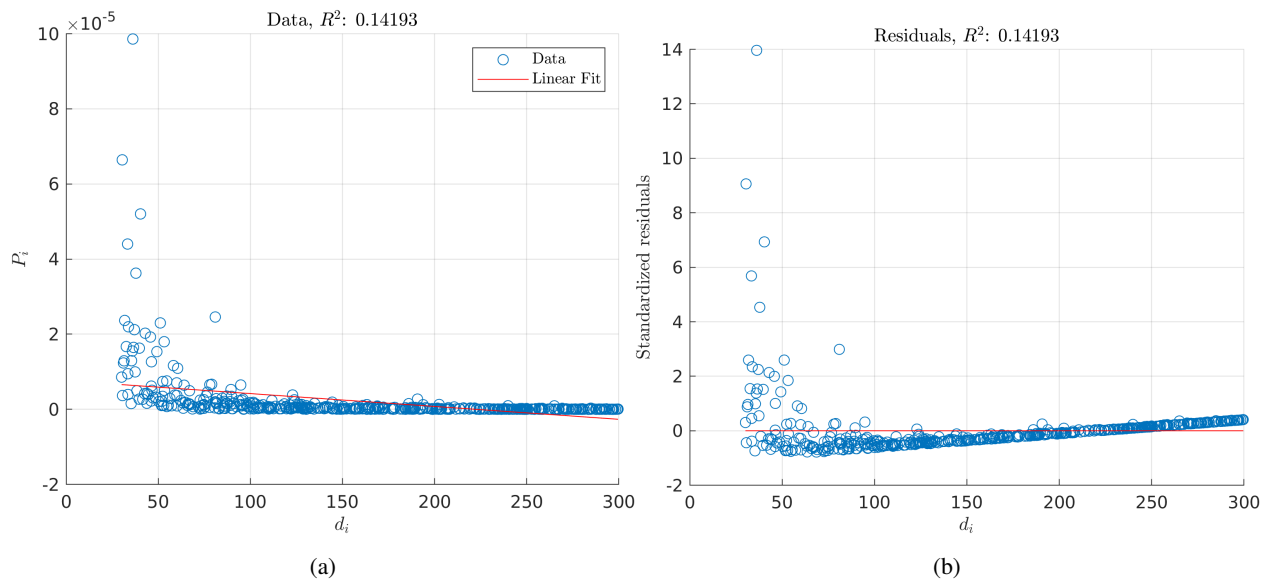


Fig. 2. Linear regression and residual analysis for raw data.

- b) From Fig. 2, $\hat{\beta}_0 = 7.61 \cdot 10^{-6}$, $\hat{\beta}_1 = -3.42 \cdot 10^{-8}$, $R^2 = 0.142$. The residuals exhibit heteroscedasticity, as the variance is higher when d_i is lower).

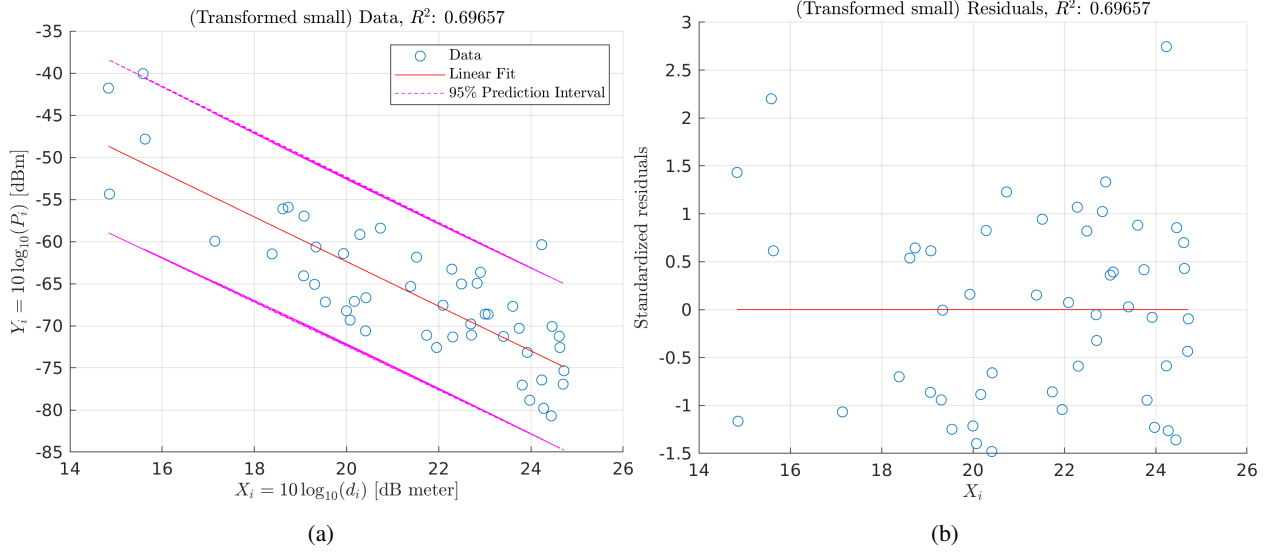


Fig. 3. Linear regression and residual analysis with $n' = 50$.

c) By applying the dB transformation on the equation, we have

$$10 \log_{10}(P_i) = 10 \log_{10}(K) - \gamma 10 \log_{10}(d_i) - 10 \log_{10}(L_i)$$

and setting

$$10 \log_{10}(P_i) = Y_i$$

$$\beta_0 = 10 \log_{10}(K)$$

$$\beta_1 = -\gamma$$

$$10 \log_{10}(d_i) = X_i$$

$$\epsilon_i = 10 \log_{10}(L_i)$$

we obtain the usual linear regression equation. Note that it is also possible to define the independent variable as $X_i = \log_{10}(d_i)$ and having $\beta_1 = -10\gamma$. The estimator $\hat{\beta}_1$ will be 10 times bigger, but nothing else will change.

d) L_i must be a RV with log-normal distribution. This would be a random variable whose logarithmic version is a Gaussian. This is how it's generated in the data, which is why everything works smoothly.

e) Fig. 3 shows the plots. $\hat{\beta}_0 = -9.3426$, $\hat{\beta}_1 = -2.6518$, $R^2 = 0.6966$.

f) Fig. 4 shows the plots. $\hat{\beta}_0 = -7.6779$, $\hat{\beta}_1 = -2.6968$, $R^2 = 0.7182$.

g) Based on the answer to c), a possible estimator is $\hat{\gamma} = -\hat{\beta}_1$, $\hat{K} = 10^{\hat{\beta}_0/10}$.

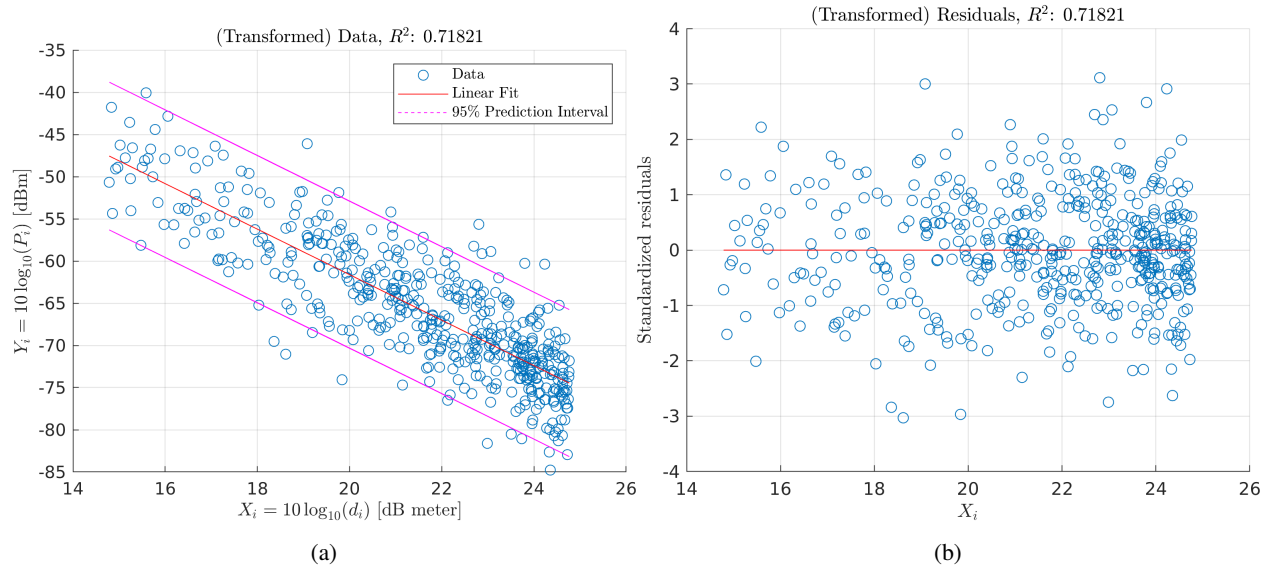


Fig. 4. Linear regression and residual analysis with $n = 500$.

- h) Using the mean response it is $Y_* = -78.3443$ with $P_{0.95} \in [-88.1317, -68.5569]$ for $n' = 50$ and $Y_* = -77.8490$ with $P_{0.95} \in [-86.392, -69.306]$ for $n = 500$. The prediction intervals are obtained implementing the formula on slide 20.